1.6.1 Some more special properties of square numbers

(i) The square of a natural number other than 1, is either a multiple of 3 or exceeds a multiple of 3 by 1.

(ii) The square of a natural number, other than 1, is either a multiple of 4 or exceeds a multiple of 4 by 1.

(iii) The remainder of a perfect square when divided by 3, is either 0 or 1 but never 2.

(iv) The remainder of a perfect square, when divided by 4, is either 0 or 1 but never 2 and 3.

(v) When a perfect square number is divided by 8, the remainder is either 0 or 1 or 4, but will never be equal to 2, 3, 5, 6 or 7.



Perfect numbers such as 6, 28, 496, 8128 etc., are not square numbers.



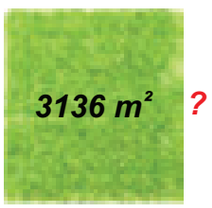
Note

If a perfect square number ends in zero, then it must end with even number of zeroes always. We can verify this for a few numbers in the table given below.

Number	10	20	30	40	...	90	100	110	...	2000	...
Its Square	100	400	900	1600	...	8100	10000	12100	...	4000000	...

Example 1.22

The area of a square field is 3136 m^2. Find its perimeter.



Solution:

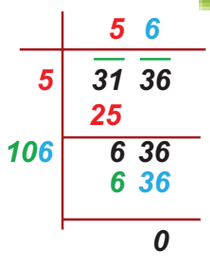
Given that the area of the square field = 3136 m^2.

$\therefore$ The side of square field = $\sqrt{3136}$ = 56 m

$\therefore$ The perimeter of the square field = 4 × side

= 4 × 56

= 224 m





Think

Consider the claim: "Between the squares of the consecutive numbers n and $(n+1)$, there are $2n$ non-square numbers". Can it be true? How many non-square numbers are there between 2500 and 2601? Verify the claim.

Example 1.23

A real estate owner had two plots, a square plot of side 39 m and a rectangular plot of dimensions 100 m length and 64 m breadth. He sells both of these plots and acquires a new square plot of the same area. What is the length of side of his new plot?

01_8th_Maths_Numbers_Chap 1.indd 27 2/12/2020 12:04:42 PM